Gonzalo N. Molina

Ph.D. student in Condensed Matter Physics

Madrid, Spain | gonzalo.molina@imdea.org | g98molina.github.io

PROFILE

Computational physicist with a B.Sc. in Physics (5-year degree) from the Universidad Nacional de Río Cuarto (Argentina) and currently a Ph.D. student in Condensed Matter Physics at the Universidad Autónoma de Madrid under the supervision of Dr. José Ángel Silva-Guillén and Dr. Francisco Guinea at IMDEA Nanociencia. My research focuses on the theoretical study of two-dimensional materials and quantum systems using first-principles methods (DFT/DFPT), with applications to topological materials, transition-metal dichalcogenides, and other low-dimensional systems.

TECHNICAL SKILLS

DFT Codes	Quantum ESPRESSO, SIESTA
Programming	Python (NumPy, Matplotlib, ASE), Mathematica, Bash
Tools	LaTeX, Git, Linux

RESEARCH EXPERIENCE

Predoctoral Researcher — 03/2024 – Present
IMDEA Nanociencia | Madrid, Spain

- First-principles modeling of quantum and two-dimensional materials using Quantum ESPRESSO and SIESTA.

Research Stay — 09/2025 – 10/2025
Departamento de Física, Universidad Nacional de Río Cuarto | Argentina

- DFT simulations of graphene and fluorinated carbon oxides for the protection of lithium metal anodes in next-generation batteries.

EDUCATION

Ph.D. in Condensed Matter Physics, Nanoscience and Biophysics — 2024 – Present
Universidad Autónoma de Madrid

B.Sc. in Physics — 2017 – 2023
Universidad Nacional de Río Cuarto, Argentina

PUBLICATIONS

"Microscopic Insights into Magnetic Warping and Time-Reversal Symmetry Breaking in Topological Surface States of Rare-Earth-Doped Bi_2Te_3." *Advanced Materials* (2025). DOI: 10.1002/adma.202510877. [*Co-author*]

CONFERENCE PRESENTATIONS

XL Reunión Bienal de la Real Sociedad Española de Física — 07/2026
First-principles calculations of magnetic defects in rare-earth-doped Bi_2Te_3 | [Poster]

2026 Sesión Científica Anual del Programa de Doctorado en Física de la Materia Condensada, Nanociencia y Biofísica — 06/2026
First-principles calculations of magnetic defects in rare-earth-doped Bi_2Te_3 | [Poster]

Artificial Intelligence for Advanced Materials (AI4AM 2026) — 05/2026
First-principles calculations of magnetic defects in rare-earth-doped Bi_2Te_3 | [Poster]

15th Early Stage Researchers Workshop in Nanoscience — 12/2025
First-principles calculations of magnetic defects in rare-earth-doped Bi_2Te_3 | [Poster]

III Workshop en Energías Renovables (UNC–UNCA–UNDEF) — 10/2025
Simulación DFT de grafeno y óxidos de carbono fluorados para la protección de litio metálico en baterías | [Poster]

110° Reunión de la Asociación Física Argentina – 3ª Reunión Conjunta AFA–SUF — 09/2025
Estudio teórico de la dispersión de fonones del hBN mediante un modelo continuo | [Oral presentation]

2025 Sesión Científica Anual del Programa de Doctorado en Física de la Materia Condensada, Nanociencia y Biofísica 05/2025
Phonon Dispersion in Hexagonal Boron Nitride: A Continuum Model Approach | [Poster]

22nd International Workshop on Computational Physics & Materials Science 01/2025
Study of the exfoliation of two-dimensional Cu-doped $TiSe_2$ | [Poster]

14th Early Stage Researchers Workshop in Nanoscience 12/2024
Phonon Dispersion in Twisted h-BN: A Continuum Model Approach | [Poster]

XXXIX Reunión Bienal de la Real Sociedad Española de Física 06/2024
Study of the exfoliation of two-dimensional Cu-doped $TiSe_2$ | [Poster]

108° Reunión Anual de Física Argentina (RAFA 2023) 09/2023
Estudio de la exfoliación de $TiSe_2$ bidimensional dopado con Cu | [Poster]

II Workshop en Energías Renovables (UNC–UNCA–UNDEF) 10/2023
Óxidos de carbono fluorados para protección de Li metálico mediante DFT | [Poster]

I SiModAr: Simposio de Modelado Multiescala para Biociencias y Nanomateriales en Argentina
11/2022
Estudio mediante cálculos de primeros principios de materiales carbonosos modificados con flúor para su utilización como protección de litio metálico | [Poster]

NANO 2022: XXI Encuentro de Superficies y Materiales Nanoestructurados 08/2022
Estudio de la exfoliación de $TiSe_2$ dopado con Cu mediante cálculos de DFT | [Poster]

SCIENCE OUTREACH

XV Feria "Madrid es Ciencia" 2026
Representative at the joint IMDEA Institutes stand, participating in nanoscience outreach activities for the general public.

Semana de la Ciencia y la Innovación 2025 2025
Workshop "Las formas del carbono: taller 3D con moléculas y efecto moiré" for children.

"Investigación Computacional en Física: DFT. Del grado en la UNRC al doctorado en Madrid." 2025
Outreach talk delivered to students and faculty members of the Departamento de Física, Universidad Nacional de Río Cuarto.

"¿Cómo funciona un mar de electrones?" 2022
Oral presentation delivered as part of the lecture series "Física: Desde Átomos Hasta Volcanes", aimed at the Facultad de Ciencias Exactas, Físico-Químicas y Naturales, Universidad Nacional de Río Cuarto.

GRADUATE COURSES

Computational Spectroscopy: Hands-on Many-Body Calculations with the Yambo Code 05/2026
MaX CoE in collaboration with NCC Czechia.

MaX School: Materials and Molecular Modelling with Quantum ESPRESSO 06/2024
MaX CoE in collaboration with NCC Czechia, NCC Austria, NCC Slovakia, NCC Slovenia, NCC Poland and NCC Hungary.

PWTK–2024: An Online Tutorial 05/2024
MaX CoE in collaboration with NCC Slovenia.

Introducción a los Métodos de Modelado Computacional en Ciencias de los Materiales 10–12/2023
Universidad Nacional de La Plata, Argentina.